

An Analysis of Expectation maximization algorithm

Expectation maximization algorithm

Expectation maximization algorithm -- Part 1

where $\hat{x}_k$ are scalar output estimates calculated by a filter or a smoother from N scalar measurements z_k . The above update can also be applied to updating a Poisson measurement noise intensity. Similarly, for a first-order auto-regressive process, an updated process noise variance estimate can be calculated by

Expectation maximization algorithm -- Part 2

The observed data points $\mathbf{X}$ may be discrete (taking values in a finite or countably infinite set) or continuous (taking values in an uncountably infinite set). Associated with each data point may be a vector of observations. The missing values (aka latent variables) $\mathbf{Z}$ are discrete, drawn from a fixed number of values, and with one latent variable per observed unit. The parameters are continuous, and are of two kinds: Parameters that are associated with all data points, and those associated with a specific value of a latent variable (i.e., associated with all data points whose corresponding latent variable has that value). However, it is possible to apply EM to other sorts of models. The motivation is as follows. If the value of the parameters $\boldsymbol{\theta}$ is known, usually the value of the latent variables $\mathbf{Z}$ can be found by maximizing the log-likelihood over all possible values of $\mathbf{Z}$, either simply by iterating over $\mathbf{Z}$ or through an algorithm such as the Viterbi algorithm for hidden Markov models. Conversely, if we know the value of the latent variables $\mathbf{Z}$, we can find an estimate of the parameters $\boldsymbol{\theta}$ fairly easily, typically by simply grouping the observed data points according to the value of the associated latent variable and averaging the values, or some function of the values, of the points in each group. This suggests an iterative algorithm, in the case where both $\boldsymbol{\theta}$ and $\mathbf{Z}$ are unknown:

Expectation maximization algorithm -- Part 3

In words, choosing $\boldsymbol{\theta}$ to improve $Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(t)})$ causes $\log p(\mathbf{X} | \boldsymbol{\theta})$ to improve at least as much.

Expectation maximization algorithm -- Part 4

The expectation of $\log L(\boldsymbol{\theta} ; \mathbf{x}_i, \mathbf{Z}_i)$ inside the sum is taken with respect to the probability density function $P(\mathbf{Z}_i | \mathbf{X}_i = \mathbf{x}_i ; \boldsymbol{\theta}^{(t)})$, which might be different for each $\mathbf{x}_i$ of the training set. Everything in the E step is known before the step is taken except $T_{j,i}$, which is computed according to the equation at the beginning of the E step section. This full conditional expectation does not need to be calculated in one step, because τ and μ/Σ appear in separate linear terms and can thus be maximized independently.

Expectation maximization algorithm -- Part 5

$$L(\boldsymbol{\theta} ; \mathbf{x}) = \prod_{i=1}^n \sum_{j=1}^2 \tau_{ji} f(\mathbf{x}_i ; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j),$$
$$\tau_{ji} = \frac{\sum_{\mathbf{Z}_i} \mathbf{1}_{\{\mathbf{Z}_i = \mathbf{z}_j\}} p(\mathbf{Z}_i | \mathbf{X}_i = \mathbf{x}_i ; \boldsymbol{\theta})}{\sum_{\mathbf{Z}_i} p(\mathbf{Z}_i | \mathbf{X}_i = \mathbf{x}_i ; \boldsymbol{\theta})}$$

Expectation maximization algorithm -- Part 6

We take the expectation over possible values of the unknown data $\mathbf{Z}$ under the current parameter estimate $\boldsymbol{\theta}^{(t)}$ by multiplying both sides by $p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{(t)})$ and summing (or integrating) over $\mathbf{Z}$. The left-hand side is the expectation of a constant, so we get:

Expectation maximization algorithm -- Part 7

$$\log p(\mathbf{X} | \boldsymbol{\theta}) - \log p(\mathbf{X} | \boldsymbol{\theta}^{(t)}) \geq Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(t)}) - Q(\boldsymbol{\theta}^{(t)} | \boldsymbol{\theta}^{(t)})$$
$$Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(t)}) = \sum_{\mathbf{Z}} p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{(t)}) \log p(\mathbf{X} | \boldsymbol{\theta}, \mathbf{Z})$$

Expectation maximization algorithm -- Part 8

where $p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta})$ is the conditional distribution of the unobserved data given the observed data $\mathbf{x}$ and D_{KL} is the Kullback-Leibler divergence. Then the steps in the EM algorithm may be viewed as:

Expectation maximization algorithm -- Part 9

However, this quantity is often intractable since $\mathbf{Z}$ is unobserved and the distribution of $\mathbf{Z}$ is unknown before attaining $\boldsymbol{\theta}$

An Analysis of Expectation maximization algorithm

Expectation maximization algorithm

$\{\boldsymbol{\theta}\}.$

Expectation maximization algorithm -- Part 10

$\log p(\mathbf{X} \mid \boldsymbol{\theta}(t)) = Q(\boldsymbol{\theta}(t) \mid \boldsymbol{\theta}(t)) + H(\boldsymbol{\theta}(t) \mid \boldsymbol{\theta}(t))$
 $, \{\displaystyle \log p(\mathbf{X} \mid \boldsymbol{\theta}^{(t)}) = Q(\boldsymbol{\theta}^{(t)} \mid \boldsymbol{\theta}^{(t)}) + H(\boldsymbol{\theta}^{(t)} \mid \boldsymbol{\theta}^{(t)})\}$

Expectation maximization algorithm -- Part 11

$L(\boldsymbol{\theta}; \mathbf{x}, \mathbf{z}) = p(\mathbf{x}, \mathbf{z} \mid \boldsymbol{\theta}) = \prod_{i=1}^n \prod_{j=1}^J [f(\mathbf{x}_i; \boldsymbol{\mu}_j, \Sigma_j) \tau_j] I(\mathbf{z}_i = j), \{\displaystyle \begin{aligned} L(\boldsymbol{\theta}; \mathbf{x}, \mathbf{z}) &= p(\mathbf{x}, \mathbf{z} \mid \boldsymbol{\theta}) \\ &= \prod_{i=1}^n \prod_{j=1}^J [f(\mathbf{x}_i; \boldsymbol{\mu}_j, \Sigma_j) \tau_j] I(\mathbf{z}_i = j) \end{aligned}$

Expectation maximization algorithm -- Part 12

These are called the "membership probabilities", which are normally considered the output of the E step (although this is not the Q function of below). This E step corresponds with setting up this function for Q:

Expectation maximization algorithm -- Part 13

Introduction The EM algorithm is used to find (local) maximum likelihood parameters of a statistical model in cases where the equations cannot be solved directly. Typically these models involve latent variables in addition to unknown parameters and known data observations. That is, either missing values exist among the data, or the model can be formulated more simply by assuming the existence of further unobserved data points. For example, a mixture model can be described more simply by assuming that each observed data point has a corresponding unobserved data point, or latent variable, specifying the mixture component to which each data point belongs. Finding a maximum likelihood solution typically requires taking the derivatives of the likelihood function with respect to all the unknown values, the parameters and the latent variables, and simultaneously solving the resulting equations. In statistical models with latent variables, this is usually impossible. Instead, the result is typically a set of interlocking equations in which the solution to the parameters requires the values of the latent variables and vice versa, but substituting one set of equations into the other produces an unsolvable equation. The EM algorithm proceeds from the observation that there is a way to solve these two sets of equations numerically. One can simply pick arbitrary values for one of the two sets of unknowns, use them to estimate

the second set, then use these new values to find a better estimate of the first set, and then keep alternating between the two until the resulting values both converge to fixed points. It's not obvious that this will work, but it can be proven in this context. Additionally, it can be proven that the derivative of the likelihood is (arbitrarily close to) zero at that point, which in turn means that the point is either a local maximum or a saddle point. In general, multiple maxima may occur, with no guarantee that the global maximum will be found. Some likelihoods also have singularities in them, i.e., nonsensical maxima. For example, one of the solutions that may be found by EM in a mixture model involves setting one of the components to have zero variance and the mean parameter for the same component to be equal to one of the data points.